

A novel thermal comfort model for older adults – development and validation of the COMFA-OA model

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ABSTRACT

This study presents a population-specific adaptation of the COMFA model, customized to estimate thermal stress levels of older adults. We refer to this index as the COMFA thermal comfort – older adults (COMFA-OA) model. This model incorporates physiological adjustments to enhance accuracy in estimating thermal stress for the older demographic. Key modifications include empirical updates to sweat heat loss, core temperature, and metabolic rate equations to reflect age-related changes in heat regulation and metabolic efficiency. The model was validated using the ASHRAE Global Thermal Comfort Database II and a field experiment assessing thermal sensation of older adults during warm and hot seasons in Texas. Results show COMFA-OA (MAE = 4.03, RMSE = 4.14) outperforms traditional models like PET (MAE = 4.72, RMSE = 4.83), UTCI (MAE = 4.82, RMSE = 4.95), and COMFA (MAE = 4.17, RMSE = 4.29) in accuracy of predicting thermal sensation vote, demonstrating greater multinomial logit model fit (AIC: 4435.3) and achieving competitive computational efficiency (TOPS = 97.54) second only to original COMFA (TOPS = 331.21) and COMFAcourtyard (TOPS = 481.87). Sensitivity analysis identified air temperature, radiant temperature, age, weight, and height as primary contributors to thermal comfort variance in older adults. The COMFA-OA model offers a practical tool for environmental management and public health applications aimed at promoting climate resilience and thermal security for older populations that are vulnerable to heat conditions.

1. Introduction

1.1. Research background

Population aging is reshaping societies worldwide. As more countries experience rapid aging, demographers have emphasized its impact on public health [1]. In 2009, approximately 39.5 million Americans were aged 65 or older; by 2050, the projected population of older Americans will reach 88.5 million, 19 million of whom will be 85 years or older [2]. This demographic shift underscores the necessity of addressing the health and well-being of older adults. One of the most important health concerns is increasingly severe extreme heat. By 2050, urban land expansion will result in average summer daytime and nighttime air temperatures warming by 0.5 °C–0.7 °C, and up to about 3 °C in certain locations [3]. Extreme heat increases the risk of multiple

morbidities, such as cardiovascular and respiratory diseases among older urban populations. Urban heat islands further exacerbate these impacts among those with existing vulnerabilities [2].

Accurately estimating thermal stress is therefore critical for public health. Thermal comfort indices vary in considerations of energy transfer process, target population, and accessibility as open-source tools. The Wet Bulb Globe Temperature index was introduced in 1954 as a “warning flag threshold,” which led to a significant reduction in heat-related illnesses in military training camps, from 53 to 10 per 10,000 trainees. Since then, more advanced energy balance equations have been developed, such as Physiological Equivalent Temperature (PET) [4], Standard Effective Temperature/Effective Temperature (SET*/ET*) [5], Predicted Mean Vote (PMV) [6], and Comfort Formula (COMFA) [7]. These models not only consider climatic factors, but also incorporate human physiological parameters to quantify the

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human-environment exchange.

The advancement of modern computational capabilities has further facilitated the development of more sophisticated multi-node models, such as the universal thermal climate index (UTCI) based on the Fiala multi-node model [8] and Berkeley models [9], which consider subdivided sections of the human body. These advanced models have been instrumental in aiding epidemiology research [10–12], guiding landscape design and urban planning [13,14], and informing policy development [10,12,15].

Among these models, COMFA has been identified by Coccolo et al. [31] as one of the most comprehensive methods available. It has been successfully adapted for diverse populations, including active adults [32,33], children [34], Asian older adults [64] and high-performance athletes [35]. COMFA has also been widely applied in a range of scenarios, including warm conditions [13,36,37], cold conditions [14], courtyard condition [36], walking behavior [37], vulnerability index development [15], athletic field assessment [38], sport and occupational medicine [39], and landscape performance [13,36,40,41]. The model's open-architecture design allows for full customization of equations and methodologies.

However, these existing models, including the COMFA model, were developed based on the physiology and human-environment heat exchanges of a middle-aged man of average build (i.e., 1.75 m, 75 kg, male). Therefore, they tend to oversimplify or overlook the biophysiological differences between older and younger adults. Prior research has highlighted significant differences in thermal stress regulation between older and younger adults, which are generally attributable to reduced sweat gland function and neurotransmitter activity, decreased metabolic efficiency due to diminished muscle mass or increased body fat, and lower cardiovascular output and blood flow [16]. Ignoring these differences may result in considerable errors when applying models to estimate thermal stress in older adults [17].

Beyond age-related differences, the structural design of thermal comfort models also matters. While dynamic models (e.g., multi-node or segmental models) can simulate detailed thermoregulatory responses over time—including core temperature accumulation, sweat rate fluctuations, and delayed physiological reactions—they require high-resolution temporal data, are computationally intensive, and often demand complex individual-level calibration. These requirements limit their applicability in large-scale or design-focused studies, especially when validating against self-reported thermal sensation data or in real outdoor monitoring scenarios. In contrast, steady-state models, although unable to capture the full thermal adaptation process, are computationally efficient and broadly compatible with most cross-sectional datasets. They provide a practical balance between accuracy and feasibility, making them well-suited for large-scale applications where high-resolution temporal data are unavailable. Their computational efficiency and simplicity also facilitate integration into routine practice, design and planning processes, and policy-making, where timely and accessible thermal comfort assessments are essential.

1.2. Review of the literature on thermal regulation in older adults

Thermal regulation in older adults differs from that in younger adults, impacting their ability to store or dissipate heat. Research has identified several underlying physiological differences in older adults, such as reduced sweat volume due to sweat gland degradation, higher metabolic cost of walking due to muscle weakness and altered muscle activation pattern, and reduced core temperature (T_c) [21].

In particular, with aging, there is overall **age-related decline in sweat production** [18] with varying degrees of local sweat rate decline with respect to passive heating. Both the output of individual sweat gland and the number of actively secreting glands decrease as aging progresses [19]. Although sweat rate and skin blood flow are regulated independently, sweat production relies on an adequate blood supply, with blood being the source of sweat liquid [20]. Reduced

cardiovascular output consequently diminishes sweat production capacity, as the increase in cardiac output associated with heat exposure is significantly smaller in older adults [19].

Reduced core temperature has also been observed in older adults [21,22]. Core temperature differences across ages can vary significantly, influenced by factors such as race [23], sex [24], measurement site (e.g., oral, rectal, axillary, or tympanic) [22], thermometer type, ambient temperature, time of day, season, and lifestyle factors [25]. Between clinically healthy adults (20–64 years) and older adult populations (65–95 years), an average temperature difference of approximately 0.4 °C has been reported, which was statistically significant [26,27]. Although the difference is physiologically mild, reduced core and skin temperatures in older adults may increase susceptibility to excessive convective heat transfer during hot summer conditions.

Metabolic heat production differs between older adults and younger adults. Older adults have both a lower resting metabolic rate and a higher net metabolic cost of walking (NCoW), due to muscle weakness, slower preferred gait speeds, and altered muscle activation patterns [28]. Consequently, older adults can have lower metabolic heat production when sitting quietly or sleeping, but higher heat production when conducting certain activities, such as moving a given distance, walking at a given speed, or lifting/lowering a given object. Existing methods mostly estimate metabolic heat based on the standard metabolic rate (METs), with the assumption that 1 MET is equal to an oxygen consumption rate (VO_2) of 3.5 ml/kg/min in all age groups. This assumption of constant VO_2 implies there is no NCoW difference between younger and older adults, which could introduce significant errors in metabolic heat estimation. Additionally, reduced heat production from shivering has been documented in older adults [29]. In cold-exposed adults, shivering thermogenesis in skeletal muscles can account for up to one-third of total heat production [30], but this contribution decreases with age [29].

In summary, the primary physiological differences between older and younger adults include (i) differential metabolic heat production, (ii) decreased sweating efficiency, and (iii) reduced core temperature. These characteristics contribute to observed differences in convective, evaporative, and radiative heat exchange between younger and older adults.

1.3. Study objectives

This study aims to develop a thermal comfort model for older adults by adapting the COMFA thermal comfort index [7]. The proposed COMFA-OA model retains the energy balance framework, while incorporating modifications to account for age-related biophysiological differences in heat transfer processes. While the model primarily focuses on adjusting the theoretical equations to reflect the physical heat transfer mechanisms of older adults, it incorporates recent empirical findings where existing theoretical equations do not fully capture these differences, resulting in a semi-empirical approach. Development of this model was anticipated to achieve several objectives:

- A) Accuracy: Provide a more accurate estimation of the biophysiological difference between younger and older adults to better estimate the thermal sensation of older adults.
- B) Universality: Maintain the universal nature of the thermal comfort model, making it comparable to other thermal indices. The COMFA-OA model uses parameters consistent with other thermal comfort models, including air temperature (T_a), mean radiant temperature (T_{mrt}), relative humidity (Rh), wind speed (W_s), standard metabolic rates (METs), age, sex, height, weight, and clothing insulation (Icl). In addition, the COMFA-OA model can produce an equivalent temperature (ET) output, enabling improved comparability with other thermal comfort indices.
- C) Generalizability: Preserve the simplicity of the original COMFA energy budget model and enhance its generalizability by

avoiding the inclusion of empirical evidence specific to particular groups, regions, or climate zones.

- D) Physical Interpretability: Maintain the capacity to produce heat flow values that are consistent with physical principles and intuitively interpretable within the context of thermodynamics.

2. Materials and methods

This study develops the COMFA-OA model, which builds on and modifies the COMFA model by incorporating equations that more accurately reflect energy exchange processes of older adults. It captures the physiology and heat transfer characteristics of older adults, such as metabolic rate, core and skin temperature, sweat response, and sweat dynamics. This section presents the key model modifications and details the approaches used to evaluate the model's performance.

The foundational COMFA-OA equation for calculating energy budget considers heat gains and heat losses:

$$\text{Budget} = M + \text{Rabs} - \text{Conv} - \text{Evap} - \text{TR}_{\text{emitted}}, \quad (1)$$

where M is the metabolic energy for heating up the body (W/m^2), Rabs is the absorbed solar and terrestrial radiation (W/m^2), Conv is the sensible convective heat exchange (W/m^2), Evap is the evaporative heat loss (W/m^2), and $\text{TR}_{\text{emitted}}$ is the emitted terrestrial radiation (W/m^2). The outcome is the total energy budget of an individual expressed in W/m^2 , which is further classified into five thermal stress categories (Eq. (1)). Optionally, the output can include a COMFA Equivalent Temperature (CET), which represents the equivalent air temperature at which a person in an indoor reference environment would experience the same physiological heat balance and thermal sensation as in the actual outdoor environment being evaluated. This reference indoor environment is characterized by mean radiant temperature set equal to air temperature ($T_{\text{mrt}} = T_a$), wind speed is fixed at 0.3 m/s at body height (≈ 0.5 m/s at the standard 10 m reference), and relative humidity is held constant at 50%. Human parameters are standardized with clothing insulation of 0.9 clo, standard metabolic rate of 2.3 MET (light activity), male sex, age 65, height 1.75 m, weight 75 kg, and standing posture.

2.1. Metabolic heat calculation for older adults (M)

Metabolic heat is determined from an age-appropriate resting metabolic rate (RMR) and the additional heat generated during physical activities. The standard metabolic rate (METs) used the original Harris-Benedict equation and assumed a fixed VO_2 consumption of 3.5 mL/kg/min, which may introduce additional error when applied to older adults. Accordingly, our model incorporates a corrected metabolic rate (MET_c) that adjusts metabolic level based on an estimate of individual RMR. In detail, the revised Harris-Benedict [43] predicted resting metabolic rate is used to correct VO_2 [42] in relation to age, height, weight, and sex (Eq. (2), Eq. (3)) [42]. This correction method was validated by Kozey et al. (2010) against indirect calorimetry, which showed that predicted MET_c derived from corrected VO_2 was statistically equivalent to MET values based on measured RMR, and more accurate than directly using 3.5 mL·kg⁻¹·min⁻¹ across diverse age, BMI, and fitness subgroups. However, this correction can overestimate RMR for sedentary individuals. Therefore, in this study, the correction is applied only to the portion of METs above 1MET (Eq. (4)), indicating that it is used solely for periods of increased metabolic activity (Eq. (4)).

$$\text{RMR} = \begin{cases} 9.65 \times \text{Weight} + 573 \times \text{Height} - 5.8 \times \text{Age} + 260, & \text{male} \\ 7.38 \times \text{Weight} + 607 \times \text{Height} - 2.31 \times \text{Age} + 43, & \text{female} \end{cases} \quad (2)$$

$$\text{VO}_{2\text{corrected}} = \frac{\text{RMR}/7200}{\text{Weight}} \times 1000 \quad (3)$$

$$\text{MET}_c = 1 + (\text{MET}_s - 1) \times \left(\frac{3.5}{\text{VO}_{2\text{corrected}}} \right), \quad (4)$$

where $\text{VO}_{2\text{corrected}}$ is the corrected oxygen consumption based on the revised Harris-Benedict RMR in mL·kg⁻¹·min⁻¹, Weight is the body weight in kg, Height is the body height in meters, RMR is calculated based on the Harris-Benedict equations as revised by Pavlidou et al. in kJ/day [43], MET_s is the standard metabolic rate in MET, and MET_c is the corrected metabolic rate in MET.

Then, the total metabolic heat production is determined as:

$$M = (1 - f) \times \left(\frac{\text{RMR}}{24} \times 1.163 \right) \times \frac{\text{MET}_c \times 58.15}{80} \quad (5)$$

where $(\text{MET}_c \times 58.15)/80$ is the activity factor defined by the original COMFA model, M is the metabolic heat production in W/m^2 , and f is a small fraction for estimating the energy consumed by respiratory latent and sensible heat losses:

$$f = 0.150 - 0.0173P_v - 0.0014 \times T_a, \quad (6)$$

where P_v is the vapor pressure estimated via the Tetens equation in Kpa, and T_a is the air temperature in °C.

2.2. Evaporative heat loss in older adults (Evap)

In hot outdoor environments where the air temperature is higher than one's skin temperature, sweat is the only means of losing heat. The total evaporative heat loss is the sum of basal and activity-triggered sweat heat loss and the invisible sweat heat loss:

$$\text{Evap} = (Q_{\text{SweatDiffusion}} + Q_{\text{SweatBasal}} + Q_{\text{SweatAct}}) \times \alpha_{\text{sweat}}, \quad (7)$$

where Evap is the total evaporative heat loss in W/m^2 , $Q_{\text{SweatDiffusion}}$ is the invisible sweat heat loss, $Q_{\text{SweatBasal}}$ is the basal visible sweat heat loss triggered by environmental factors, Q_{SweatAct} is the visible sweat heat loss triggered by metabolism or high-intensity exercise, and α_{sweat} is the coefficient to adjust for sweat degradation with aging.

Invisible sweat heat loss. The invisible sweat heat loss is calculated as:

$$Q_{\text{SweatDiffusion}} = (1.16 \times 2442) \times \left(\frac{q_s - P_v}{7700 + R_{cv} + 0.92 \times R_a} \right), \quad (8)$$

where q_s is the saturation-specific humidity at skin temperature, P_v is the vapor pressure in kPa, R_{cv} is the resistance of clothing to water vapor in s/m, and R_a is the aerodynamic resistance in s/m.

Activity-triggered sweat heat loss. Activity-triggered sweat heat loss is modeled as a function of metabolic rate, consistent with the approach used in COMFA:

$$Q_{\text{SweatAct}} = 0.42 \times (M - 58.15), \quad (9)$$

where M is the metabolic heat in W/m^2 , Q_{SweatAct} is the activity-triggered sweat in W/m^2 .

Environment-triggered sweat heat loss. The COMFA model applies the ISO 7730 standard and assumes sweat evaporation is based solely on metabolic rate (Eq. (9)). Considering that older adults typically have lower activity levels, relying solely on Q_{SweatAct} while ignoring resting-state sweat may lead to errors in estimating their sweat heat loss. To address this, we introduce a new basal sweat heat loss estimation ($Q_{\text{SweatBasal}}$) to account for the sweat levels of older adults in environments with reduced physical activity. Any thermal stress input, from external stimulation (climatic exposure) to activity (activity metabolic rate) and RMR, can potentially trigger basal sweat. Omidvar and Kim suggested that the value of basal sweat heat loss should be estimated from the predicted values of sweating rates at higher activity levels [44]. Our estimation of basal sweat hat loss employs Omidvar and Kim's [44]

extended application of a fuzzy piecewise prediction equation function (PW function) [45], where they proposed low-activity sweat estimation by setting $M = 4$ in the original PW formula and incorporated a β coefficient to address overestimation in low-sweat scenarios (sweat < 200 g/m²/hour) (Eq. (10), 11, 12):

$$Q_{\text{SweatBasal}} = \frac{0.725}{1 + 0.155I_{\text{cl}}(4.6 + h_c + 0.046T_{\text{mrt}})} \times M^*_{\text{swb}} \times \beta, \quad (10)$$

where I_{cl} is clothes insulation in clo, h_c is the convective heat transfer coefficient, T_{mrt} is the mean radiant temperature in °C, β is the sweat level adjustment, and M^*_{swb} is the piecewise linear prediction of the basal sweating rate (when METs = 4 MET) [44,45].

M^*_{swb} is calculated as:

$$M^*_{\text{swb}} = (36.75 + 0.3818E_{\text{req(METs=4)}} - 0.2175E_{\text{max(METs=4)}}), \quad (11)$$

where $E_{\text{req(METs=4)}}$ is the net energy required to reach energy balance when METs = 4 MET and $E_{\text{max(METs=4)}}$ is the maximum evaporation limited by environmental and human body parameters when METs = 4 MET.

$$\beta = \begin{cases} 1 & \text{if } 200 \leq M^*_{\text{swb}}, \\ 0.5 & \text{if } 150 \leq M^*_{\text{swb}} < 200, \\ 0.33 & \text{if } M^*_{\text{swb}} < 150 \end{cases} \quad (12)$$

Age-related sweat degradation. COMFA-OA additionally applies an age-scaling coefficient to account for the progressive decline in sweat gland activity after early adulthood. While the grouped-age design of key studies (e.g., Larose et al., 2013 [46]) does not resolve the exact functional form by single year, decade-scale trends are compatible with a near-linear attenuation across adulthood [47,63]. Given the lack of a universally established nonlinear function across adult ages and the moderate sample sizes available, COMFA-OA represents this effect using a first-order, monotonic linear scaling for adults aged ≥ 20 y:

$$\alpha_{\text{sweat}} = \begin{cases} 1 - (\text{age} - 20) \times 0.005, & \text{age} \geq 20 \\ 1, & \text{age} < 20 \end{cases} \quad (13)$$

where 0.005 is the age-related sweat coefficient for adults 20 years or older. This linear scaling is intended as a population-level approximation within a steady-state energy-budget framework, not a claim of strict linearity in all contexts. Future work with richer covariates (fitness, acclimation, medication, comorbidities) and repeated-measures designs could test nonlinear or piecewise forms (e.g., decade-specific slopes).

2.3. Radiation heat gain (Rabs) and radiation heat loss (TR_{emitted})

Rabs and TR_{emitted} can be calculated based on measured T_{mrt} and the radiation heat transfer equation. Given possible wheelchair uses by older adults [48], we also applied a body posture coefficient in the calculation of heat exchange between the human body and its environment:

$$R_{\text{abs}} = (T_{\text{mrt}} + 273.15)^4 \times (k_B \times 0.97) \times \beta_{\text{rad,eff}} \quad (14)$$

$$TR_{\text{emitted}} = (T_{\text{sk}} + 273.15)^4 \times (k_B \times 0.95) \times \beta_{\text{rad,eff}} \quad (15)$$

$$\beta_{\text{rad,eff}} = \begin{cases} 0.895 & \text{if standing} \\ 0.691 & \text{if sitting} \\ 0.600 & \text{if crouching} \end{cases}, \quad (16)$$

where T_{mrt} is the mean radiant temperature in °C, k_B is the Boltzmann constant, and $\beta_{\text{rad,eff}}$ is the radiation heat transfer coefficient, with values of 0.895 for standing, 0.691 for sitting, and 0.600 for crouching.

2.4. Convective heat loss in older adults (Conv)

Convective heat loss is estimated through the convective heat

transfer equation, which considers the exposed body surface area and that covered by clothes:

$$\text{Conv} = - \frac{Q_{\text{convbare}} + Q_{\text{convclo}}}{BSA} \quad (17)$$

$$Q_{\text{convbare}} = H_{\text{conv}} \times (T_a - T_{\text{sk}}) \times S_{\text{nude}} \quad (18)$$

$$Q_{\text{convclo}} = H_{\text{conv}} \times (T_a - T_{\text{sf}}) \times S_{\text{clo}} \quad (19)$$

$$\begin{aligned} h_{\text{c,free}} &= 2.38(T_{\text{sf}} - T_a)^{0.25}, \\ h_{\text{c,forced}} &= (2.67 + 6.5V^{0.67}) \left(\frac{p_{\text{Baro}}}{p_{\text{setting}}} \right)^{0.55}, \\ H_{\text{conv}} &= \max(h_{\text{c,free}}, h_{\text{c,forced}}) \end{aligned} \quad (20)$$

where Q_{convbare} is the convective heat loss of bare body surface area in W/m², Q_{convclo} is the convective heat loss of clothed surface area in W/m², BSA is the total body surface area in m², H_{conv} is the convective heat transfer coefficient, S_{nude} & S_{clo} are area of bare and clothed body area in m², T_a is the ambient temperature in °C, T_{sk} is the skin temperature in °C, T_{sf} is the clothes surface temperature in °C, and Vr is the relative speed between human subject and wind in m/s, p_{baro} is the constant standard barometric pressure used for normalization, set to 1013.25 hPa, p_{setting} is the the measured barometric pressure at the site in hPa.

In the original COMFA model, Vr is simplified as the square root of the sum of the human activity speed (V_{ac}) and wind speed (W_s). Notably, this determination might underestimate the relative air flow when either V_{ac} or V_a is extremely low, which can result in significant error when an older adult is moving slowly. In the COMFA-OA model, Vr is modeled as the mathematical expectation of wind speed randomly comes from all directions (0 to 360 degrees):

$$V_r = \frac{1}{2\pi} \int_0^{2\pi} \sqrt{V_{\text{ac}}^2 + W_s^2 - 2 \times V_{\text{ac}} \times W_s \times \cos(\theta)} d\theta \quad (21)$$

$$V_{\text{ac}} = 0.0052 \times (58.15 \times \text{MET}_s - 58.15 \times 1.5), \quad (22)$$

where V_{ac} is the activity speed in m/s and W_s is the wind speed in m/s.

Age-related core temperature differences in older adults. Older adults tend to have lower core temperatures compared to younger individuals, with reported average reductions of up to 0.7 °C. This reduced core temperature is partially reflected in reduced metabolic heat production, although a 10 % reduction in metabolism accounts for a core temperature drop of only 0.025 °C [49], suggesting need for a distinct adjustment of the core temperature representation. In our COMFA-OA model, core temperature is modeled as a function of metabolism with adjustment based on a linear function developed by Protsiv et al. [50], as their study featured one of the largest sample sizes and careful control of body height and weight. The correction coefficient β_{age} is respectively set for males and females as 0.0060 and 0.0066 [50]:

$$T_c = \begin{cases} (36.5 + (0.0043 \times M)) - \beta_{\text{age}} \times (\text{Age} - 20), & \text{if Age} \geq 20 \\ 36.5 + (0.0043 \times M), & \text{if Age} < 20, \end{cases} \quad (23)$$

where

$$\beta_{\text{age}} = \begin{cases} 0.0060, & \text{for male} \\ 0.0066, & \text{for female} \end{cases} \quad (24)$$

Estimation of mean skin temperature. In COMFA-OA, mean skin temperature (T_{sk}) is revised to account for more age-related factors, including relative speed, core temperature, and metabolic heat. In addition, unlike the traditional model that solely considers the convective heat transfer resulting from air flow over the skin boundary layer,

COMFA-OA incorporates radiation heat, convective heat, and metabolic heat transfer in a new empirical formula derived based on a large experimental study [51]. This results in a set of three equations for three different clothing scenarios:

For fully clothed subjects ($I_{cl} \geq 0.6$):

$$\begin{aligned} \bar{T}_{sk} = & 12.2 + 0.020 \times T_a + 0.044 \times T_{mrt} + 0.194 \times P_v \\ & - 0.253 \times V_r + \\ & 0.00297 \times MET_c + 0.513 \times T_c \end{aligned} \quad (25)$$

For minimally clothed subjects ($I_{cl} \leq 0.2$):

$$\begin{aligned} \bar{T}_{sk} = & 7.2 + 0.064 \times T_a + 0.061 \times T_{mrt} + 0.198 \times P_v - 0.348 \\ & \times V_r + \\ & 0.616 \times T_c \end{aligned} \quad (26)$$

For intermediate clothing conditions ($0.2 \leq I_{cl} \leq 0.6$):

$$\bar{T}_{sk} = \bar{T}_{sk_{nude}} + 2.5 \times (\bar{T}_{sk_{clothed}} - \bar{T}_{sk_{nude}}) \times (I_{cl} - 0.2), \quad (27)$$

where $\bar{T}_{sk}$ is the average skin temperature in $^{\circ}\text{C}$, P_v is the vapor pressure in Kpa, T_a is the air temperature in $^{\circ}\text{C}$, T_{mrt} is the mean radiant temperature in $^{\circ}\text{C}$, MET_c is the corrected metabolic heat production in MET, and T_c is the core temperature in $^{\circ}\text{C}$.

Convective heat model based on ratio of exposed and clothed body surface. Original COMFA model assumes that the skin is entirely covered by clothes, leading to errors convective heat transfer. COMFA-OA improves the precision of convective heat transfer estimation by considering the ratio of exposed to clothed body surface area [52]. While this adjustment does not directly address physiological differences between younger and older adults, it enhances the accuracy of convective heat transfer estimations related to revised age-related parameters such as mean skin temperature and core temperature.

$$f_{clo} = \frac{(-2.36 + 173.51 \times I_{cl} - 100.76 \times I_{cl}^2 + 19.28 \times I_{cl}^3)}{100}, \quad (28)$$

where f_{clo} represents the proportion of the body covered by clothing (range 0–1) and I_{cl} is the thermal insulation due to clothing ($1 \text{ clo} = 0.155 \text{ K}\cdot\text{m}^2/\text{W}$).

Additionally, Spastic's clothing expansion formula (Eq. (28)) was incorporated to estimate clothing expansion, the contribution of garment thickness and thermal insulation to a larger surface area. This adjustment improves the model's accuracy in representing real-world convective heat transfer [53].

$$f_{expansion} = -1 + (0.31 \times I_{cl}) \quad (29)$$

$$S_{clo} = BSA \times f_{clo} + BSA \times (f_{expansion} - 1.0) \quad (30)$$

$$S_{nude} = BSA \times (1 - f_{clo}), \quad (31)$$

where $f_{expansion}$ is the clothes expansion coefficient, S_{clo} is the clothed area in m^2 , and S_{nude} is the unclothed body area in m^2 . Then, the convective heat transfer for exposed and clothed body parts can respectively be written as:

$$Q_{ConvBare} = -h_{conv} \times (T_a - \bar{T}_{sk}) \times S_{nude} \quad (32)$$

$$Q_{ConvClo} = -h_{conv} \times (T_a - T_{cl}) \times S_{clo} \quad (33)$$

where $\bar{T}_{sk}$ is the average skin temperature in $^{\circ}\text{C}$, T_{cl} is the clothes surface temperature in $^{\circ}\text{C}$.

Finally, the overall convective heat transfer is calculated as the sum of the convection of the exposed and clothed body surface areas:

$$\text{Conv} = -\frac{(Q_{ConvBare} + Q_{ConvClo})}{BSA}, \quad (34)$$

where Conv is the convective heat loss in W/m^2 , $Q_{ConvBare}$ is the convective heat gain in exposed skin, and $Q_{ConvClo}$ is the convective heat gain for clothed areas.

2.5. Evaluation of model performance

Two datasets were used for the evaluation of model performance. The data characteristics and variables are described as follows.

Dataset I: ASHRAE Global Thermal Comfort Database II. The ASHRAE Global Thermal Comfort Database II (ASHRAE II TSV) [54] was released in 2018 and includes data from thermal comfort studies published in the 1990s to 2010s representing diverse climates, environmental conditions, and cultures. This dataset has strict inclusion and validation criteria, which ensure data quality. From this dataset, we selected sample points with complete input parameters (T_a , R_h , W_s , I_{cl} , sex, age, weight, height, METs, and globe temperature $[T_g]$) for participants over 50 years old, resulting in a total of 1487 data points. Thermal sensation vote (TSV) was also extracted, categorized into seven rounded and resampled levels: cold (−3), cool (−2), slightly cool (−1), neutral (0), slightly warm (1), warm (2), and hot (3). Notably, the ASHRAE II dataset is limited by a lack of high-temperature data; the majority of data points for older adults ($N = 1007$) fall within a T_a range of 20–30 $^{\circ}\text{C}$, with about a tenth as many representing higher temperatures ($T_a \geq 35$ $^{\circ}\text{C}$, $N = 110$).

Dataset II: Outdoor Thermal Comfort Survey for Older Adults. To ensure good sample coverage across the full scope of ambient temperature and age ranges, we designed a field experiment on the Texas A&M University campus. First, outdoor microclimate conditions across campus were monitored by the research team with an ambulatory weather station MaxiMet GMX501 between August and December 2024. Based on this data, the team selected twelve locations representing two levels of wind conditions (mild and high wind) and two levels of solar radiation (shade and no shade), with three locations for each combination. In the experiment, each participant was invited to two sessions on separate days, and were randomly assigned to experience one of the twelve sites for each session (Fig. 1).

We recruited participants through senior centers and organizations in town, sending flyers and postcards to homes in residential areas, events, the campus bulk email system, and snowball methods. The inclusion criteria consisted of 1) age 55 and above, 2) able to walk to the designated location on campus, 3) does not have uncontrolled coronary heart disease, stroke, or hypohidrosis that impacts daily activities, 4) can speak English or Spanish and 5) give consent to participating in the study.

The experiment was conducted during daytime hours, from 9:00 to 16:30. Participants were instructed to avoid coffee and iced drinks for at least three hours prior to the experiment. After checking in and giving informed consent, participants spent 15 min in a temperature-controlled room and filled out background questionnaires. This served as a conditioning phase to ensure participants were thermally comfortable and remove the effects of differing travel conditions on the way to the study site. Then, the participants were randomly assigned to one of the twelve locations on campus, where they were instructed to remain seated for 15 min. During that period, they answered questions about their thermal comfort every five minutes, measured using the 9-point TSV questionnaire (Supplementary Material Figure S1), which ranges from cold (−4) to neutral (0) and then to hot (4). In analyses, we only used the third measurement taken at 15 min, considering it to best reflect the impacts of the outdoor spot microclimate condition. In total, each session lasted an average of 50 min. Participants received a stipend of \$80 for completing two sessions, as well as a complimentary parking pass. During each experiment, we collected environmental data, including

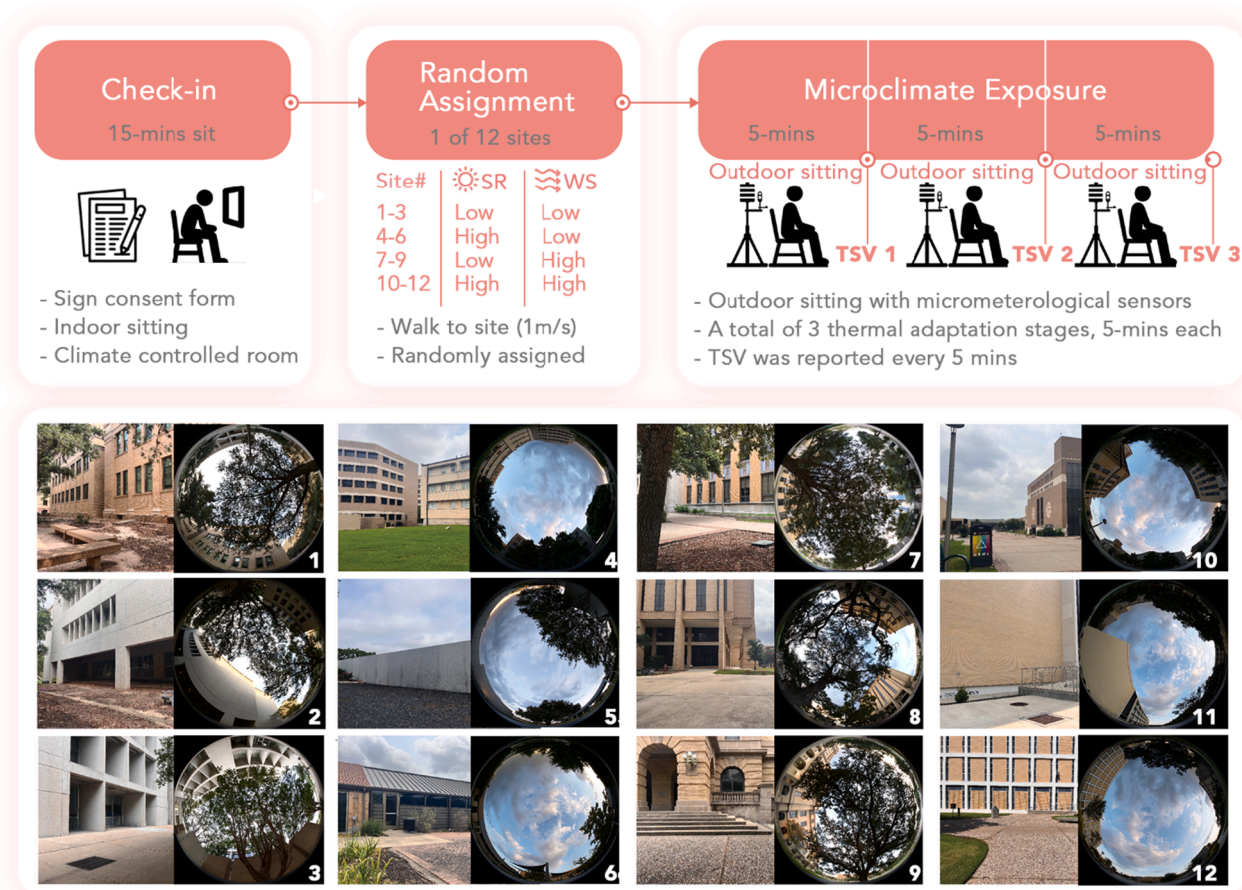


Fig. 1. Experimental procedure and 12 sites involved.

direct shortwave radiation, black globe temperature, Ws, Ta, Rh, and barometric pressure, using MaxiMet GMX501 and Kestrel 5400 sensors installed right next to the participant, at 1.5 m above ground level

Table 1
Specifications of microclimate sensors used in the study.

Variables	Sensor name	Country of origin	Interval	Accuracy
Ta*	Maximet GMX501	Gill Instruments, UK	10-sec	± 0.3 °C
Sr*	Maximet GMX501	Gill Instruments, UK	10-sec	± 5 % (ISO 9060 2nd Class)
Ws*	Maximet GMX501	Gill Instruments, UK	10-sec	± 0.3 m/s (RMSE)
Rh*	Maximet GMX501	Gill Instruments, UK	10-sec	± 1 %
Pa*	Maximet GMX501	Gill Instruments, UK	10-sec	Absolute ± 0.4 hPa Relative ± 0.08 hPa
Ta ⁺	Kestrel 5400	Nielsen-Kellerman, USA	10-sec	± 0.5 °C
Tg*	Kestrel 5400	Nielsen-Kellerman, USA	10-sec	± 1.4 °C
Ws ⁺	Kestrel 5400	Nielsen-Kellerman, USA	10-sec	Larger of 0.1 m/s or ± 3 %
Rh ⁺	Kestrel 5400	Nielsen-Kellerman, USA	10-sec	± 2 %
Pa*	Kestrel 5400	Nielsen-Kellerman, USA	10-sec	Absolute ± 1.5 hPa

* First choice.

⁺ Alternative choice

Ta: Air temperature. Tg: Black globe temperature. Ws: wind speed. Rh: relative humidity. Pa: Barometric pressure. Sr: Shortwave solar radiation.

(Table 1). Data collection occurred in two rounds, with 235 participants in the first round and 177 returning for the second round, yielding a total of 412 TSV data points (collectively referred to as TSV-OA hereafter).

Model performance comparison. In the data analysis phase, the ASHRAE TSV and TSV-OA datasets were merged into a single dataset and the distribution of TSV values was visualized as a stacked histogram.

To comparatively evaluate performance of the COMFA-OA model, we calculated other thermal comfort outcomes using COMFA, COMFA-courtyard, UTCI, PET, SET* and ET* ([36,52]) models in Python. A multinomial logit regression model (MNL) was applied to assess the extent to which each resulting thermal index predicted TSV [57,58]. The MNL approach is commonly used to model relationships between continuous predictors and ordinal categorical outcomes, especially when the proportional odds assumption was violated. Additionally, a random effect is incorporated into the MNL model to account for potential within-subject repeated measures, as some participants completed two rounds of data collection. Model fit was quantified in terms of the Akaike Information Criterion (AIC) and McFadden's pseudo- R^2 .

Thermal stress models typically classify continuous thermal comfort values into 5 or 7 ordinal categories representing levels of thermal stress. Accordingly, we followed established approaches to determine category thresholds [36], applying an ordinary least squares (OLS) model to describe the relation between the energy budget value and the mean thermal sensation vote within a fixed bin width (TSVm). Specifically, we created 5 $W \cdot m^{-2}$ bins of the index (analogous to 1 °C bins used with PET/UTCI), computed the mean TSV (TSVm) within each bin, and fit an ordinary least-squares line $TSVm \sim \text{Index}$ [36]. For each bin, the average TSV was defined as the corresponding TSVm. To reduce

instability from sparse tails, bins with fewer than three observations were excluded. We explicitly note that TSV is ordinal. MNL model therefore underpins our predictive/inferential evaluation, whereas OLS-by-bins is used only as a calibration mapping to derive interpretable cut-points when categories are unbalanced (e.g., few very cold observations) and ordered-model cut points can become unstable. This quasi-interval treatment of TSV at the bin-mean level is long-standing in the field and standard practices [58] and serves to denoise individual-level response variability. Resulting thresholds should thus be interpreted as practical calibration boundaries rather than strict probabilistic decision thresholds

To further quantify predictive accuracy, we calculated the mean absolute error (MAE) and root mean square error (RMSE) for COMFA, COMFA-OA and other top three models with lowest AIC, providing complementary perspectives on average prediction deviation and

sensitivity to large errors.

Additionally, as thermal stress assessment may be applied in large datasets, we assessed the computational speed of the COMFA-OA calculation in thousands of operations per second (TOPS) [36]. We further compared its performance with other models in terms of execution time on a personal computer equipped with an Intel i7-9700k CPU operating at 3.6 GHz.

2.6. Sensitivity analysis

To determine how changes in model inputs affect COMFA-OA sweat level estimation and evaluate the model's robustness against extreme values, we conducted a sensitivity analysis following the methodology of Kenny et al. (2009) [33]. Presets for all parameters except T_a , T_{mrt} , and W_s were determined based on the average values of the validation

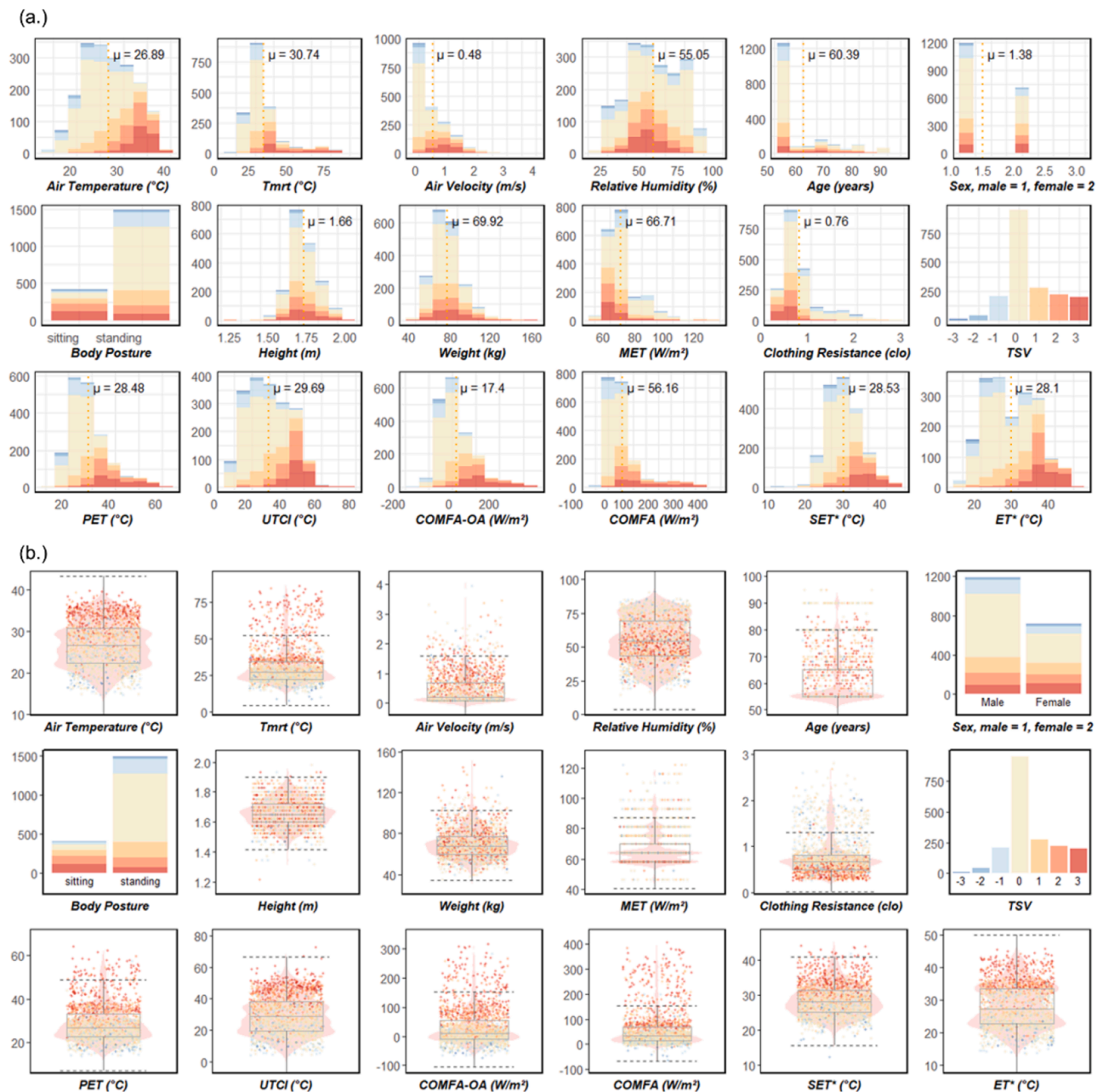


Fig. 2. (a) Distribution of TSV values for the validation dataset in relation to input parameters; colors in stacked histograms indicate TSV category. (b). Distribution of TSV values for the validation dataset in box plot and violin plot, point colors correspond to TSV category. Box plot whiskers extending to $Q1 - 1.5 \text{ IQR}$ and $Q3 + 1.5 \text{ IQR}$.

dataset. The hot and cold conditions respectively corresponded to the 90th and 10th percentiles of T_a and T_{mrt} . W_s was a consistent 2 m/s for both hot and cold conditions. Finally, the values of each target variable were independently varied by a set increment within an appropriate range (Supplementary Material Table S1).

We further performed Sobol sensitivity analysis with a sampling number of 5000 using the “sensitivity” package in R [59] to evaluate the first order and total-order sensitivity of COMFA-OA, along with other thermal comfort models. Supplementary Material Table S2 presents the variable ranges used in the Sobol analysis.

3. Results

3.1. Data distribution

The validation dataset (Fig. 2) includes COMFA-OA and other comparable thermal comfort indices, along with a range of micrometeorological and physiological parameters. Overall, the sample provides good coverage for T_a , T_{mrt} , W_s , and R_h , while emphasizing warmer conditions for accurate characterization of heat stress. In terms of participant demographics, there was a slightly higher proportion of males, and the age distribution was primarily centered between 50 and 60 years, with a small secondary peak observed around 70–80 years. MET was centered around 66 W/m², likely due to the low activity level of the older adult participants. Clothing insulation showed a single peak and a mean value of 0.76, implying that light clothing was most commonly worn. For TSV categories, neutral (0) was by far the most common, with warm and hot conditions predominating over extreme cold. All thermal comfort indices showed a unimodal distribution except ET*, which exhibited bimodality.

In addition to stacked histograms, Fig. 2b presents boxplots for each continuous variable, showing quartile ranges, medians, and whiskers extending to $Q1 - 1.5$ IQR and $Q3 + 1.5$ IQR. Individual data points are overlaid and color-coded by TSV. Climatic outliers were evident in high T_{mrt} and high wind speed, capturing some extremely hot and windy conditions. No notable outliers were found for T_a or R_h , consistent with their relative stability. For human parameters, outliers appeared at higher weights and older ages, indicating the inclusion of some overweight and “older-old” participants. The clo values above 1.3 were identified as outliers, consistent with the predominantly summer sampling period. While most participants were sedentary, a few cases with light activity levels were also flagged as outliers.

3.2. Model performance

In predicting thermal sensation of older adults, the COMFA-OA model demonstrated superior accuracy and reliability, outperforming the other five models examined (Table 2). Specifically, COMFA-OA achieved the lowest AIC values in the MNL, at 4435.3, indicating a strong model fit. The second-lowest AIC was obtained with the PET model, at 4453.4. The UTCI, ET*, and COMFA models also demonstrated moderately good performance.

The MNL further identified the breakpoint for neutral thermal

sensation (TSV = 0) in each model. For COMFA-OA, the peak of the TSV = 0 curve corresponded to an energy value of -15.2 W/m² (Fig. 3); in contrast, the peak for the COMFA model occurred at 9.1 W/m². These results suggest that older adults prefer a lower energy budget.

In addition to statistical fit, we assessed the computational efficiency of the various models, which revealed COMFA-OA to offer competitive calculation speed (97.54 TOPS). Although slower than COMFA (331.207 TOPS) and COMFAcourtyard (481.874 TOPS), it executed significantly faster than the other thermal comfort indices tested. In addition, COMFA-OA is the only model to provide both physically meaningful energy flux and the equivalent temperature; other extant models give either energy flux (COMFA) or the equivalent temperature (PET, UTCI, SET*, ET*).

3.3. Model sensitivity

Sobol sensitivity. The Sobol sensitivity analysis identified two climatic factors, T_a ($S_T = 0.63$) and T_{mrt} ($S_T = 0.20$), as the major contributors in the COMFA-OA model (Supplementary Material Figure S2), with W_s ($S_T = 0.07$) and R_h ($S_T = 0.002$) being less impactful. Notably, the COMFA model was highly sensitive to T_{mrt} but less sensitive to human parameters, whereas the COMFA-OA model was much more sensitive to subject age ($S_T = 0.08$), body height ($S_T = 0.07$), body weight ($S_T = 0.14$), and sex ($S_T = 0.06$) (Supplementary Material Figure S2).

Parameter sensitivity. The impacts of individual model parameters on COMFA-OA results were assessed and plotted (Supplementary Material Figure S3). Results show that METs and R_h have strong impacts on net energy budget, especially under hot conditions. In contrast, parameters like I_{cl} and W_s have more pronounced effects in cold conditions.

Energy flux sensitivity. The individual COMFA-OA energy flux (see Eq. 1) change with the increasing of major input variable were plotted in Supplementary Material Figure S4. We decomposed Eq. (1) to show how individual heat-flux terms respond to the major inputs (Supplementary Material Fig. S4). T_{mrt} dominates radiation heat gain; metabolic heat gain increases with standard MET; convective heat loss is strengthened by higher W_s and larger skin/clothing-air temperature gradients (especially in cold presets), with I_{cl} and W_s more impactful in cold; Evaporative heat loss is the primary regulator in hot conditions and is maximized by low I_{cl} , higher W_s , low R_h , and higher MET; emitted radiation heat is comparatively stable. These component-level responses explain how changes in inputs propagate to the COMFA-OA net energy budget and the resulting thermal-stress category.

3.4. Model interpretive thresholds

Thermal stress thresholds were selected following established methods, specifically ordinary regression with increments of energy budget values. The OLS model identified -13.97 and -35.27 W/m² for COMFA-OA and COMFA, respectively, as the breakpoints for thermally neutral sensation (Fig. 4).

The respective fitted regression equations were written as:

Table 2
Comparison of thermal comfort models and their performance.

Thermal comfort index	Model input	Model output	Calculation speed (TOPS)	Multinomial Logit Regression (MNL) AIC	R ² (McFadden)
COMFA-OA	Climatic factors/human parameters	Energy flux / ET	97.54	4435.291	0.219
COMFA	Select climatic factors/human parameters	Energy flux	331.207	4715.103	0.169
PET	Climatic factors/human parameters	ET	2.69	4453.404	0.215
UTCI	Climatic factors	ET	74.80	4500.391	0.207
SET*	Climatic factors/human parameters	ET	7.70	4717.832	0.168
ET*	Climatic factors/human parameters	ET	7.53	4525.397	0.203
COMFAcourtyard	Select climatic factors/human parameters	Energy flux	481.874	4623.221	0.185

ET: Equivalent temperature. TOPS: Thousand operations per second. MNL: Multinomial logit regression model.

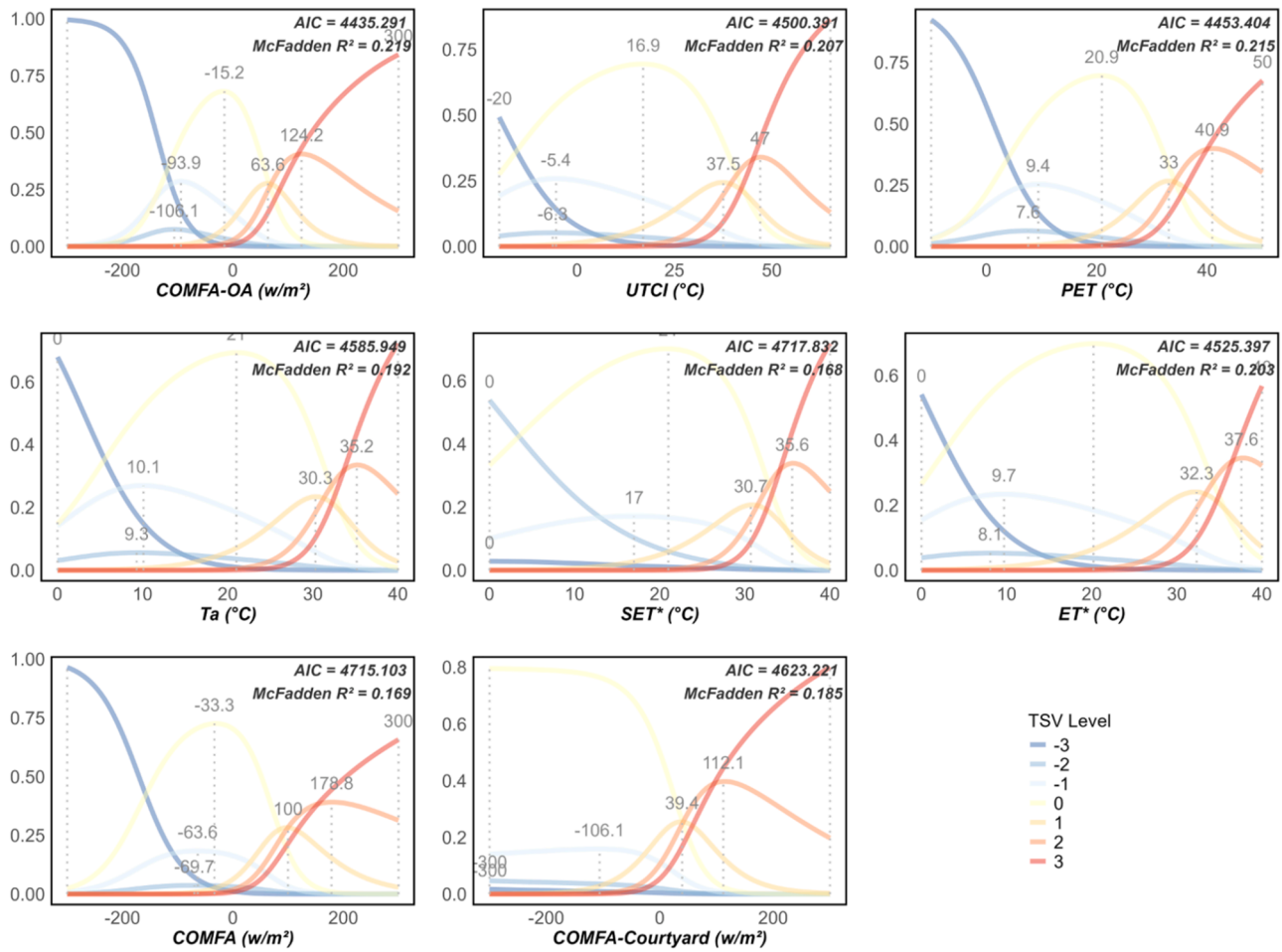


Fig. 3. Multinomial logit (random intercept): Predicted category probabilities of TSV across indices. The y-axis shows predicted probability; model-fit metrics are listed at top-right of each panel.

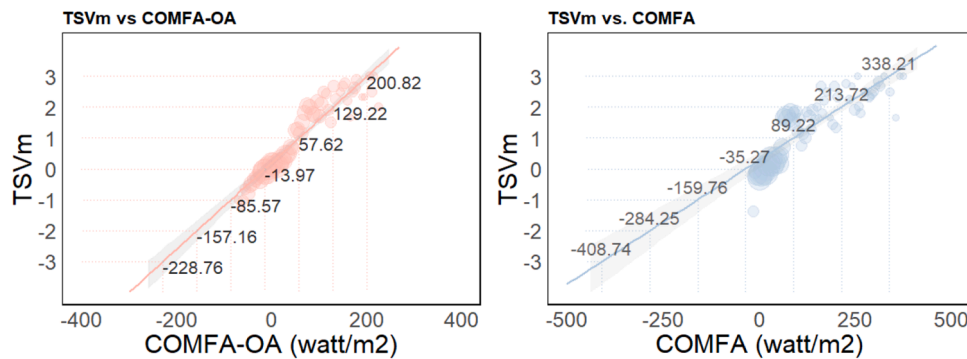


Fig. 4. Linear regression results relating TSVm to energy budget values from the COMFA-OA and COMFA models. A bigger cluster indicates a larger amount of sample points within the bin. A total of seven energy budget values were labeled in the fitted line, representing the energy budget values corresponding to TSVm = -3, -2, -1, 0, 1, 2, and 3.

$$\text{TSVm} = 0.1941 + 0.0139 \times \text{COMFA}_{\text{OA}} \quad (R^2 = 0.8885) \quad (35)$$

$$\text{TSVm} = 0.2833 + 0.0080 \times \text{COMFA} \quad (R^2 = 0.7199) \quad (36)$$

where TSVm is the mean thermal sensation vote in each bin and COMFA and COMFA_{OA} denote energy budget values in W/m². These equations were used to calculate upper and lower limits for each thermal sensation category (Table 3).

Both COMFA-OA and COMFA models identify the neutral thermal

sensation as occurring within a slightly negative energy budget, but with different ranges. Specifically, COMFA-OA defines the neutral thermal sensation range as -49 to 22 W/m², while COMFA defines it as -97 to 27 W/m². The narrower thresholds of neutral and moderate categories with COMFA-OA reflect the greater sensitivity of older adults to extreme microclimate conditions. For example, the threshold for very hot (+3) is 165 W/m² in COMFA-OA but 276 W/m² with COMFA. Similarly, the slope of the regression line for COMFA-OA ($k = 0.0139$) is steeper than that for COMFA ($k = 0.0080$), indicating that for a given change in

Table 3

Thermal sensation category thresholds based on predicted energy budget values.

Predicted thermal sensation	COMFA-OA (W/m^2)	COMFA (W/m^2)
-3 (Cold)	< -192	< -346
-2 (Cool)	-192 to -121	-346 to -222
-1 (Slightly cool)	-120 to -50	-221 to -98
0 (Neutral)	-49 to 22	-97 to 27
1 (Slightly warm)	23 to 93	28 to 151
2 (Warm)	94 to 165	152 to 276
3 (Hot)	> 165	> 276

energy flux, older adults experienced a more pronounced impact on thermal sensation than do younger adults.

Accuracy of the COMFA-OA and COMFA models was assessed through a confusion matrix comparing predicted thermal sensation level with actual TSV (Fig. 5). COMFA-OA demonstrated a higher overall accuracy rate ($n = 989$, 52.13 % across categories -3 to +3) compared to COMFA (43.12 %). For the neutral condition (0), COMFA-OA accurately predicted 80.2 % of cases, whereas COMFA accurately predicted only 58.3 %. COMFA-OA also outperformed COMFA in all hot categories (+2 and +3), correctly predicting 25.5 % of hot (+2) and 23.6 % of very hot (+3) cases, versus corresponding rates of 14.7 % and 21.2 % for COMFA. However, for the warm condition (+1), COMFA achieved higher accuracy, at 69.5 % versus 40.7 %. Both models performed poorly for the slightly cool condition (-1), with COMFA-OA predicting 5.3 % and COMFA failing to predict any cases correctly. When the prediction standard was relaxed to include outputs within ± 1 point as successful, COMFA-OA's accuracy rose to 86.5 % for slightly cool and 7.5 % for cold conditions, significantly outperforming the corresponding 69.7 % and 0 % accuracy achieved by COMFA.

3.5. False positives/negatives in hot conditions

We defined a false negative as a case where the observed thermal sensation vote (TSV = +3, very hot) was predicted by the model at least two units lower ($PTS - TSV \leq -2$). Conversely, a false positive was defined as a case where the model predicted very hot ($PTS = +3$), but the participant reported a TSV at least two units lower ($PTS - TSV \geq 2$). Using the confusion matrix (Fig. 5), we extracted these cases for further examination.

A total of 86 false negative cases and only one false positive case were identified. The single false positive occurred at 3:00 pm during an extremely hot period ($T_a \approx 36^\circ C$, $T_g \approx 52^\circ C$, shortwave radiation $\approx 1000 W/m^2$). In this case, all indices predicted "very hot" ($PTS = +3$), while the participant reported otherwise, suggesting a likely self-reporting error. For the 86 false negative cases, the distribution is shown in Fig. 6 (a - d). These cases were concentrated around $T_a \approx 35^\circ C$ and $T_{mrt} \approx 35^\circ C$. Notably, 45.1 % of the false negative cases were reported by female participants ($N = 39$) compared with a 37.5 % female representation in the full validation dataset ($N = 71$; see Fig. 2).

Kernel density contours indicated three high-frequency false negative zones (Fig. 6d). Zones #1 and #2 were characterized by high T_a and

medium T_{mrt} , located along the linear T_a - T_{mrt} trend line that reflects stable indoor radiation conditions. These zones suggest that thermal discomfort driven primarily by high air temperature in stable radiative environments tends to be underestimated by COMFA-OA in both sexes. In contrast, Zone #3 was characterized by medium T_a ($\sim 31^\circ C$) and elevated T_{mrt} ($\sim 40^\circ C$), representing outdoor environments with fluctuating radiation. All false negative cases in Zone #3 were female, indicating that COMFA-OA may underestimate heat stress in outdoor radiant conditions, particularly for women.

4. Discussion

4.1. Major modifications to COMFA model

In the COMFA-OA model, the decline in core temperature for older adults was quantified using an age-related function. In addition, the model separately evaluated elements of sweat production, dividing it into basal and activity-triggered components, rather than relying solely on activity-triggered sweat. This approach allows for more accurate assessment in individuals with low activity levels, such as older adults with limited mobility. Regarding parameter contributions to the model, age is on par with height and second only to body weight in its influence. Additionally, the metabolic heat production component was adjusted to account for variations in metabolic efficiency, indirectly addressing the effects of muscle degradation and reduced cardiac output among older adults. Finally, COMFA-OA is able to output not only energy flux but also equivalent temperature, a widely utilized concept whose incorporation enhances the model's flexibility and comparability.

In summary, the COMFA-OA model addresses age-related differences in physiological parameters and energy transfer processes in order to better reflect the characteristics of older adults. The model design prioritized corrections to physical energy transfer processes, with empirical formulas applied where the basic energy budget model could not fully account for observed variations. The optional equivalent temperature output further ensures model comparability.

4.2. COMFA-OA performance and sensitivity

Among the indices evaluated, COMFA-OA provides the best overall fit for estimating thermal stress in older adults while preserving computational efficiency due to its simple energy-budget formula structure [36]. This efficiency makes COMFA-OA particularly suitable for high-resolution datasets, real-time processing, and large-scale thermal-comfort assessments. PET and UTCI are slightly less precise but remain to be acceptable alternatives given their broad application in earlier studies [31,55,56]. COMFA-OA is also the only model that outputs both physically explicit energy-flux components and an equivalent temperature, enabling detailed diagnosis for designers and intuitive communication with non-specialists. Sensitivity tests showed that the age-specific physiological refinements in COMFA-OA significantly improved the model's responsiveness to variations in age, sex, height, and weight. However, the false negative analysis revealed that

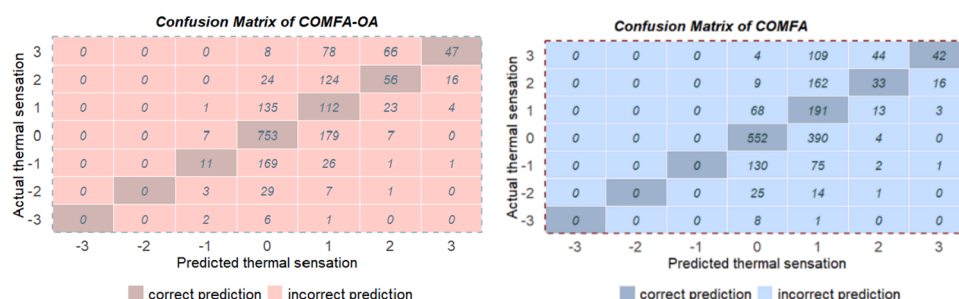


Fig. 5. Confusion matrices of predicted versus actual TSV for COMFA-OA and COMFA models.

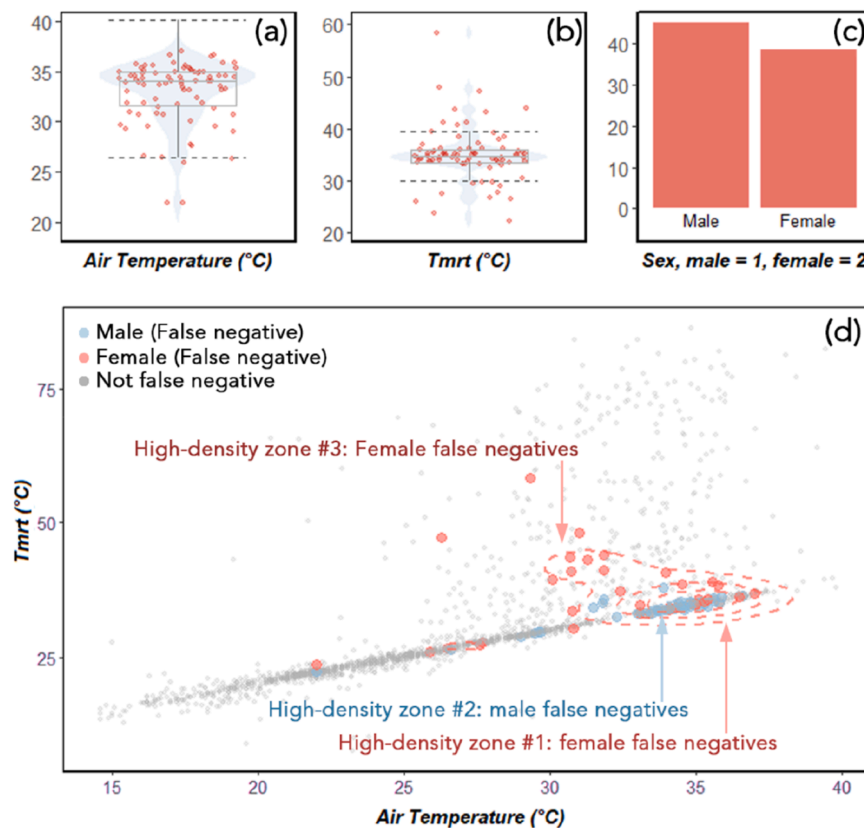


Fig. 6. (a - c) False negative cases, subset with TSV = 3, and PTS - TSV ≤ -2. (a.) air temperature distribution of false negative cases; (b.) Mean radiant temperature distribution of false negative cases; (c.) sex of false negative cases; (d.) Scatter points of full validation dataset, with false negative cases highlighted by larger, colored markers. Kernel density contours illustrate the concentration of male (blue dashed lines) and female (red dashed lines) cases.

COMFA-OA may underpredict extreme heat stress under conditions of elevated air temperature with stable radiation (for both sexes) and in high-radiation outdoor environments (particularly among females). This underscores the need for further refinement in sex-specific and temperature-radiation-interaction calibrations.

4.3. COMFA-OA predictive power for older adults

Direct comparison of predictive errors showed COMFA-OA had the smallest MAE (4.03) and RMSE (4.14), followed by COMFA original (MAE = 4.17, RMSE = 4.29), PET (MAE = 4.72, RMSE = 4.83), and UTCI (MAE = 4.82, RMSE = 4.95). The Friedman rank sum test indicated significant differences in prediction error among the four models ($\chi^2 = 1922$, $df = 3$, $p < 0.001$). Post-hoc pairwise comparisons revealed that COMFA-OA significantly increased the predictive accuracy compared to PET, UTCI, and COMFA original (all $p < 0.001$), whereas PET and UTCI showed no significant difference ($p = 0.96$).

Compared to the original COMFA energy balance model, COMFA-OA demonstrated greater overall accuracy in older adults, with a 21 % relative improvement in successful predictions (52 % vs. 43 %, +9 percentage points). Moreover, it achieved high accuracy (89.7 %) within a ± 1 TSV tolerance range and demonstrated superior accuracy in the cool, neutral, hot, and very hot TSV categories. In two cold conditions (cold, -2; and very cold, -3), data was insufficient.

Compared to other reports focused on specific participant populations, the new COMFA-OA model provides acceptable thermal sensation estimation. When predicting TSV in active athletic subjects, Vanos previously reported an accuracy of 48 %, which increased to 93 % when allowing ± 1 TSV tolerance [35]; these values are respectively slightly lower and slightly higher than the accuracy of COMFA-OA in older adults. Similarly, COMFA-OA demonstrated slightly higher

correlation ($r = 0.693$, $p < 0.001$) than that reported by Kenny et al. ($r = 0.574$, $p < 0.01$) for subjects performing physical activity [33]. Notably, an overall accuracy of 93.26 % ($r = 0.794$, $p < 0.01$) was reported for application of the COMFA model to a group of children [34]; however, the present work employed different statistical methods and approaches to determine model threshold, making it difficult to directly compare predictive power.

4.4. Thermal adaptivity of older adults

The COMFA-OA model effectively captures the heightened vulnerability of older adults to extreme temperatures. With a steeper fitted line slope and narrower interval between neutral (0) and very hot (3) or very cold (-3) conditions than obtained with COMFA, our model suggests that each unit of energy budget change has a greater impact on the thermal sensation of older adults. In other words, the COMFA-OA model identifies older adults as a vulnerable group whose lower thermal adaptive capacity makes them more prone to heat or cold stress. However, further studies are needed to strengthen this conclusion, particularly under cooler conditions and across different age groups.

The neutral breakpoint value identified by the multinomial logit model further highlights the unique thermal adaptation of older adults. That is, older adults seem to prefer a cooler space than general population, as the COMFA-OA model shows a neutral breakpoint of -15.2 W/m^2 , lower than the 65 W/m^2 reported for subjects performing physical activity [33] and 48 W/m^2 reported for general population in courtyard setting. Similar results have been reported when using other thermal comfort indices. For example, PET in this older adult study exhibited a lower breakpoint of 23.9°C , similar to the value reported by Du et al. (21.9°C) [60] but distinct from the higher breakpoint reported for the general population in hot-humid regions (25.5°C) [61]. Likewise, the

thermally neutral breakpoints obtained for UTCI and SET* in previous study (22.9 °C and 25.5 °C respectively) are lower than the typical values reported for the general population in hot-humid regions (24.1 °C and 26.95 °C respectively) [61].

4.5. Model applications and practices

COMFA-OA model incorporates several features that enhance its applicability in real-world decision-making. Its energy-budget framework produces both physically meaningful energy fluxes and an equivalent temperature output, facilitating integration into diverse workflows ranging from urban climate modeling to public health risk communication. The model's relatively simple structure and moderate computational demand enable its use at high spatial and temporal resolutions, as well as its embedding within real-time monitoring platforms. Furthermore, its reliance on widely measurable microclimatic and personal parameters supports deployment in contexts where specialized instrumentation is limited, making it suitable for large-scale heat-risk mapping and scenario testing in urban planning.

The model's improved fit for older adults also carries tangible public health implications. At a metropolitan scale, such an improvement could translate into thousands of additional older adults being correctly identified as at risk during extreme heat events, enabling timely and targeted interventions such as shade provision, cooling center access, or hydration reminders. In resource-limited contexts, this precision targeting can improve both the efficiency and equity of interventions, aligning with precision public health principles. Moreover, a more accurate model can enhance the fidelity of risk assessments, guiding strategic investments in heat-mitigation infrastructure—such as tree canopy expansion, reflective materials, or water features—that have been shown to reduce heat exposure and lower heat-related morbidity and mortality. In sum, COMFA-OA's combination of physiological specificity, interpretability, and operational efficiency positions it as a potential practical tool for climate-resilient urban design and heat-health preparedness for vulnerable populations.

4.6. Limitations and future directions

In this study, a variety of microclimatic conditions were sampled on multiple days from early spring to early autumn 2024 to ensure diversity of test conditions. This collected data compensated for the limited coverage of the global TSV dataset with regard to age and T_a . However, more responses were collected under warm and hot conditions (>20 °C) than cold conditions. While this provides more confidence in the model's performance in warm and hot weather conditions, when more attention is merited due to higher risk of heat-related illness [62], future research should explore a broader range of weather conditions, especially cold weather, to enhance the model's wide applicability. Another concern is the participants selection criteria of dataset II (Thermal Comfort Survey for older adult), where we conducted a larger and representative sampling, while did not consider medication and activity level. Future studies should incorporate medication-related variables to enhance internal validity. In addition, limited sample points in dataset II may lead to Data-dependence and calibration bias. The core energy-budget equations and age-specific parameters in COMFA-OA were derived from prior physiology rather than trained on ASHRAE II. However, the calibration layer that maps continuous energy budget to ordinal TSV (binning + multinomial logit) inevitably reflects the marginal distribution of the datasets used. Because ASHRAE II contributes most observations and its older-adult subset is sparser at the very hot tail (e.g., $T_a \geq 35$ °C), neutral points and hot-category thresholds may be biased toward ASHRAE II, consistent with our observed false-negative concentration under extreme heat. Although our field dataset extends hot-condition coverage, this risk remains. To strengthen external validity, future work will conduct leave-dataset-out and study-wise blocked cross-validation, provide an open re-calibration interface so

users can re-estimate thresholds with local samples, and report per-dataset thresholds and calibration curves.

For model structures, both the COMFA and COMFA-OA models are steady-state models that identify values at a single point in time and consider an average human body; they do not consider dynamic heat change over the course of a period of time, especially the core heat accumulated during exercise. Limited by model structure and existing empirical evidence, age-scaling factor for sweat is approximately as a first-order, monotonic approximation based on available grouped-age evidence. While biologically plausible deviations from strict linearity exist at finer scales, especially due to regional sweat gland heterogeneity and workload-dependent onset lags, our choice balances parsimony, identifiability, and model stability. Future versions could explore nonlinear or piecewise-linear age functions when supported by expanded datasets.

For validation method, this study only compared models in terms of net energy budget on account of length limits; future work needs to similarly validate the COMFA-OA equivalent temperature output. In addition, threshold values were determined based on the ASHRAE II and collected TSV-OA datasets; while this work is notably the first to provide specific thresholds for energy budget modeling of older adults, threshold values can vary wildly based on parameter settings and sampling coverage, and more region/climate zone/weather-specific studies are needed to determine thresholds in specific conditions. Future work should re-derive thresholds with ordered regression or probabilistic approaches when larger, more balanced samples are available. Finally, although subjective thermal sensation vote (TSV) is the most commonly used metric for validating thermal comfort models, it cannot fully capture the biophysiological response, such as indirect metabolic heat exchange, core temperature / skin temperature change and respiration rate change experienced by individuals. Future research should incorporate independent physiological outputs, such as skin temperature, clothing surface temperature, and sweat level, to provide a more comprehensive validation of thermal comfort models. Lastly, due to dataset scope and measurement constraints, this article presents preliminary validation of model predictive performance rather than a comprehensive explanation of thermal adaptation in older adults. Future studies should examine behavioral factors (e.g., clothing) and physiological sensitivity using age-controlled designs and enhanced measurements, with particular attention to specific energy-flux components and thermal environmental parameters.

5. Conclusions

To better facilitate thermal comfort analysis and research in older adults, it is necessary to develop an accurate, interpretable thermal comfort index specific for that group. This study developed the **COMFA-OA** model, based on COMFA, with consideration of the physiological differences between older and younger adults, incorporating the most recent empirical evidence along with the advantageous aspects of PET and SET* in calculating radiation, convection, and thermal conductance. The efficacy of the model was validated in four respects: 1). validation in a real scenario (**Outdoor Thermal Comfort Survey for Older Adults**) and an open-source dataset (**ASHRAE II Global Dataset**), 2). comparison with relevant models (PET, SET*, ET*, UTCI, COMFA), 3). sensitivity analysis, and 4). statistical quantification of performance. Additionally, this study identified the correspondence of predicted energy budget values with the 7-point thermal sensation votes of older adults. The main conclusions of the study are as follows:

- The COMFA-OA model achieved the best overall fit among the six models tested (PET, SET*, ET*, UTCI, COMFA), with the lowest AIC (4435.3) and the highest McFadden pseudo- R^2 (0.219). Compared to original COMFA, it also achieved improved predictive accuracy for older adults across all TSV categories. PET ranked second to COMFA-OA, showing strong adaptability for the older adult population. UTCI

also demonstrated remarkable predictive power, which is particularly valuable given the small number of model inputs it requires.

- The COMFA-OA model is highly interpretable, producing intuitive equivalent temperature and physically meaningful thermal flux values. It retains the comprehensive nature of the original COMFA model by thoroughly simulating the various thermal fluxes involved in human energy balance. By adjusting the scope of energy balance calculations, the model theoretically aligns more closely with the physiological characteristics of older adults. Its energy flux outputs are demonstrably more accurate for older adults than those of the original COMFA model. By preserving the physical meaning of each energy flux, the COMFA-OA model has the potential to inform the design of outdoor spaces tailored to the older adult population.
- The COMFA-OA model also maintains the interpretability of simple energy balance models, in contrast to those that simulate heat stress across multiple body nodes. This feature ensures high computational efficiency and enables broader applications, such as scenarios requiring refined-resolution data, real-time calculation, or large-scale spatial-level thermal comfort analysis.
- Attention is needed for high Ta-Low Tmrt scenario, where COMFA-OA may underpredict the "very hot" sensations when air temperature is high but mean radiant temperature is moderate/stable.
- In outdoor settings with rapidly fluctuating solar radiation (patchy sun/shade), COMFA-OA shows a higher false-negative rate for female participants, suggesting the need for sex-specific calibration under high-radiation variability.
- Older adults attain thermal neutrality at cooler conditions than the general population (i.e., a slightly negative energy budget), indicating that environments serving older adults should incorporate stronger radiant shielding and enhanced convective/evaporative cooling (e.g., shade canopies/trees, low-emissivity and high-albedo surfaces, seating in breezeways or with fans, and ready access to water/misting).
- We recommend running COMFA-OA with locally measured Ta/Tmrt/Ws/Rh (spot audits at ~1.5 m) to map older-adult risk, then iteratively test shading/greening/material/fan-misting scenarios and choose the option that shifts predicted TSV toward neutral.
- We recommend using COMFA-OA's energy flux diagnostics to pick mechanisms that match the deficit: increase radiant shielding (trees, canopies), improve convection (breezeways/fans, oriented seating), and support evaporation (near-at-hand water/misting and drinking access).

Ethics approval/declarations

This research has been approved by the Texas A&M Human Research Protection Program (Approval ID STUDY2024-0071).

Code availability

The code of the model algorithm is available on GitHub at [<https://github.com/UrbanHeatLabXLI/COMFA-OA-model>]. A Shiny application demonstrating the core functionality is available at [https://comfa.shinyapps.io/10_shinyapp/]. Model threshold calibration code will be provided in the GitHub repository at [<https://github.com/UrbanHeatLabXLI/COMFA-OA-model>].

CRedit authorship contribution statement

Xiaoyu Li: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yue Zhang:** Writing – review & editing, Writing – original draft, Data curation, Conceptualization. **Huiyan Sang:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Chanam Lee:** Writing – review & editing, Writing – original draft,

Methodology, Conceptualization. **William C. Sullivan:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Jay E. Maddock:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Dongying Li:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Robert D. Brown:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Dongying Li, Huiyan Sang, Chanam Lee, William C. Sullivan, Jay Maddock, Robert D. Brown reports financial support was provided by National Institute on Minority Health and Health Disparities. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.buildenv.2025.113758.

Data availability

The authors do not have permission to share data.

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